

Jinkyoun Hwangbo

 [hbjk](#) | hbjk0305@gmail.com | [hbjk0305.github.io](https://github.com/hbjk0305) |

EDUCATION

- **University of Illinois Urbana-Champaign** Aug 2026– May 2028 (Expected)
Master's Student in Computer Science Illinois, USA
- **Seoul National University** Mar 2018–Aug 2022
Bachelor of Science in Industrial Engineering and Artificial Intelligence Seoul, Republic of Korea
 - Honor: Summa Cum Laude

WORK EXPERIENCE

- **Samsung Electronics** Oct 2025–Jun 2026
Full Time Engineer @ Advanced Audio Lab, Mobile eXperience Business Suwon, Republic of Korea
 - Focus: Audio-Visual Separation, Target Speaker Extraction
- **Samsung Electronics** Jul 2022–Sep 2025
Full Time Researcher @ Language & Voice Team, Samsung Research Seoul, Republic of Korea
 - Focus: Audio-Visual Separation, On-device Sound Event Detection, Anomaly Detection
- **Samsung Electronics** Summer 2021
Research Intern @ Language & Voice Team, Samsung Research Seoul, Republic of Korea
 - Focus: Neural Bandwidth Extension
- **Seoul National University** Summer 2020
Research Intern @ Human Centered Computing Lab Gwanggyo, Republic of Korea
 - Focus: ECG-based Heart Disease Prediction

PATENTS

- [P.2] Jinkyoun Hwangbo, Hosang Sung, Seong-Hu Kim, et al. "Electronic Device and Method for Summarizing Video Data". KR Patent Application, P2025-0128738. Filed: Sep 10, 2025.
- [P.1] JeongHoon Lee, Yeaseul Song, Jinkyoun Hwangbo, et al. "Electronic Device and Method for Generating Summarized Video Data". KR Patent Application, P2025-0115037. Filed: Aug 19, 2025.

PROJECTS

- **Audio Eraser for Galaxy S series** Mar 2024–Jun 2026
Main AI-powered feature of Galaxy S series that can remove unwanted background sounds Samsung Electronics 
 - Designed lightweight, low-latency neural networks for multi-label sound event detection and optimized C++ engine for edge devices.
 - Developed a multimodal approach to identify and extract video highlights through audio-visual integration. [P.1, P.2]
- **Anomaly Detection for Samsung Home Appliances** Oct 2022–Dec 2024
Acoustic anomaly detection for home appliances, deployed in SmartThings home appliances Samsung Electronics
 - Developed data-efficient fault detection algorithm using <4 hours of training data, delivering \$1.3M in estimated annual cost reduction.
- **Speech Bandwidth Extension with Hifi-GAN** Summer 2021
Bachelor's Thesis in Artificial Intelligence (Advisors: Chanwoo Kim, Corporate EVP; Prof. Sungjoo Yoo) Samsung Electronics
 - Developed generative model extending audio bandwidth from 8 kHz to 16 kHz while restoring codec artifacts, improving MUSHRA perceptual quality scores from 66.3 to 87.1 and demonstrating generalization across unseen speakers and diverse codec conditions.
- **GAN based Sign Language Synthesis Model** May–Dec 2021
Undergraduate Research Program (Advisor: Prof. Kyomin Jung) Seoul National University
 - Led team of 3 researchers, developing GAN model that generates sign language motion video while preserving speaker facial features, securing undergraduate research grant.
- **ECG-based Heart Disease Prediction Using Deep Learning** Summer 2020
Research Internship Project at HCC Lab (Advisor: Prof. Bongwon Suh) Seoul National University
 - Developed Convolutional Recurrent Neural Network (CRNN) model for mobile heart disease prediction.

HONORS AND AWARDS

- **Samsung Research Award** (\$ 2,200 USD)
Awarded project biannually for exceptional technical achievement across global research centers. Dec 2024
- **Merit-based Scholarship (Top 5%) @ Seoul National University**
full amount (3 semesters), 50% (3 semesters), and 10% (1 semester) of tuition waiver 2019–2022
- **Reporter Award @ Seoul National University** Nov 2019
- **4th Supercomputer Idea Contest** (1st place, \$ 500 USD) Oct 2016
- **3rd Supercomputer Idea Contest** (1st place, \$ 700 USD) Oct 2015

TEACHING / MENTORING EXPERIENCE

- **Teaching Assistant** Mar 2021–Jun 2022
Courses: *Human Factors Engineering (406.304)*, *Scientific Management (406.211)*
 - Assisted students in achieving completion of homework, projects and evaluated their work
- **Volunteer Social Service**
 - SNU Mentoring Program: Served online mentoring for high school students. Mar–Nov 2021
 - Teaching assistant for young adults with developmental disabilities. Spring 2019

ACTIVITIES

- **Tobigs: Machine Learning Study Group** Jan 2021–Jan 2022
 - Smootify [🔗]: Developed system for seamless song transitions, enabling continuous infinite streaming using music generation techniques with MuseGAN.
 - MPTI [🔗]: Developed chat service that allows users to interact with movie characters using DialoGPT and GPT-2.
- **Seoul National University News [🌐]** Apr 2018–Dec 2019
 - Head of Photography Department (1 semester) – Led department and managed visual content.
 - Head of New Media Team (1 semester) – Led newly established New Media team
 - Photojournalist (2 semesters) – Worked as photographer for university newspaper.

SKILLS

- **Tools and Frameworks:** PyTorch, TensorFlow, Tensorflow Lite, NumPy, Pandas, Matplotlib, SciPy, scikit-learn
- **Programming Languages:** Python, C/C++, Java, Kotlin, MySQL
- **Relevant Courseworks:** *Graduate level:* Machine Learning, Convex Optimization, Social Computing;
Undergraduate level: Artificial Intelligence (ENG), Basics of Deep Learning, Computational Linguistics
- **Others:** Git, Github, Unit Testing, OOP, CI/CD